

1. Find whether the following statements are true or false? If $A \subseteq B$, $B \neq \emptyset$ then

- (A.) $A \cap B = A$
- (B.) $A \Delta B = B \setminus A$
- (C.) $A \cap B^c \subseteq A$.
- (D.) $A \cup B = A$

TRUE:	A	B	C		FALSE:	D			
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2. Find whether the following statements are true or false?

- (A.) $(A \cap B \cap C) \cup (A \cap B \cap C^c) \cup (A^c \cap B \cap C) \cup (A^c \cap B \cap C^c) = A$.
- (B.) $A \times (B \cap C^c) = (A \times B) \cap (A \times C^c)^c$.
- (C.) If $A \cup B = A \cup C$ and $A^c \cup B = A^c \cup C$ then $C \subseteq B$.
- (D.) If $A \Delta B = C$ then $B = C \Delta A$ and $B \cup C \subseteq A$.

TRUE:	B	C			FALSE:	A	D		
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3. For $n = 1, 2, \dots$, if $I_n = (-\frac{1}{n}, \frac{1}{n})$ and $J_n = [-1 + \frac{1}{n}, 1 - \frac{1}{n}]$ Then

- (A.) $I_n \cap J_n$ is closed interval for some $n \geq 3$.
- (B.) $\cap_{n=1}^{\infty} I_n$ is an open interval or $\cup_{n=1}^{\infty} J_n$ is an open interval.
- (C.) $(\cup_{n=1}^{\infty} J_n) \setminus (\cap_{n=1}^{\infty} I_n)$ is a closed set.
- (D.) $\cup_{n=1}^{\infty} (I_n^c \cap J_n)$ is an open set.

TRUE:	B			D	FALSE:	A	C		
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4. There exist functions $f, g : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

- (A.) g is not injective but $g \circ f$ is injective.
- (B.) f is not injective but $g \circ f$ is injective.
- (C.) g is not surjective but $g \circ f$ is surjective.
- (D.) f is not surjective but $g \circ f$ is surjective.

TRUE:	A	D			FALSE:	B	C		
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5. Let $\{u_n\}, \{v_n\}$ be any sequence of real numbers. Then

- (A.) $\{|u_n|\}$ converges to $|u|$ implies $\{u_n\}$ converges to u .
- (B.) if $u_n, v_n > 0, \forall n$ and $\{u_n\}$ converges to 0 and $\{v_n\}$ is bounded then $\{u_n \cdot v_n\}$ is convergent.

- (C.) if $\{|u_n|\}$ is Cauchy and $v_n > 0$ for all n with $|u_n - v_n| < .01$, then $\{v_n\}$ is also Cauchy.
- (D.) There exist convergent sequences $\{u_n\}, \{v_n\}$ such that $u_{2n} = v_{2n}$ (all even terms of the sequences are same) for all n and $\lim_{n \rightarrow \infty} u_n \neq \lim_{n \rightarrow \infty} v_n$.

TRUE:	B				FALSE:	A	C	D	
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6. Find which one of these are true and which are false

- (A.) For $0 < a < b < \infty$, $(a^{1/n} + b^{1/n})^n$ converges to b .
- (B.) If $\{u_n\} \rightarrow u$ and $\{v_n\} \rightarrow v$ then $\{(u_n)^{v_n}\} \rightarrow u^v$.
- (C.) $\frac{3^{4n}}{4^{3n}} \rightarrow 0$ as $n \rightarrow \infty$
- (D.) For any sequence $\{u_n\}$, the sequence $x_n := \min\{u_n, u_{n+1}, u_{n+2}, \dots\}$ is a monotonically increasing sequence.

TRUE:	D				FALSE:	A	B	C	
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7. A sequence $\{u_n\}$ is defined by $u_1 > 0$ and $u_{n+1} = \sqrt{6 + u_n}$ for $n \geq 1$. Then

- (A.) $\{u_n\}$ is monotonically increasing if $0 < u_1 < 2$.
- (B.) $\{u_n\}$ is monotonically decreasing if $u_1 > 2$.
- (C.) $\{u_n\}$ is convergent if $0 < u_1 < 2$.
- (D.) $\{u_n\}$ is divergent if $u_1 > 2$.

TRUE:	A	C			FALSE:	B	D		
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8. Let $\{u_n\} \rightarrow l > 0$ be any convergent sequence of positive numbers. Then

- (A.) $\sum_{n=1}^{\infty} \frac{u_n}{n^2}$ always converges.
- (B.) $\sum_{n=1}^{\infty} \frac{u_n}{n}$ can converge for some of such sequences.
- (C.) $\sum_{n=1}^{\infty} (u_n)^n$ converges if $l < 1$.
- (D.) $\sum_{n=1}^{\infty} (u_n)^{1/n}$ converges if $l < 1$.

TRUE:	A	C			FALSE:	B	D		
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9. Find whether the following statements are true or false?

- (A.) If $\lim_{x \rightarrow 0+} f(x)$ and $\lim_{x \rightarrow 0-} f(x)$ exist then $\lim_{x \rightarrow 0} f(x)$ exist.
- (B.) $\lim_{x \rightarrow 0} \sin x \cos \frac{1}{x}$ does not exist.

- (C.) if $f : [-1, 1] \rightarrow \mathbb{R}$ is given by $f(x) = \begin{cases} x & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$, then for any $a \in [-1, 1]$, $\lim_{x \rightarrow a} f(x)$ does not exist.
- (D.) If $f : (0, 1) \rightarrow \mathbb{R}$ such that $f(x) > 0$ for all $x \in (0, 1)$ then $\lim_{x \rightarrow 0} f(x) \geq 0$.

TRUE:					FALSE:	A	B	C	D
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10. The following functions are continuous on their respective domains:

- (A.) $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ defined by $f(x) = e^{1/x}$.
- (B.) $f : [0, 4] \rightarrow \mathbb{R}$ defined as $f(x) = \begin{cases} \frac{x^2-4}{x-2} & \text{if } x \neq 2 \\ 5 & \text{if } x = 2 \end{cases}$.
- (C.) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f = h \circ g$ where $g(x) = \text{sgn}(x)$ and $h(x) = x(1 - x^2)$ on $\mathbb{R}$.
- (D.) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f = g \circ h$ where $g(x) = \text{sgn}(x)$ and $h(x) = x(1 - x^2)$ on $\mathbb{R}$.

TRUE:	A	C			FALSE:	B	D		
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11. Let $f : (0, 1) \rightarrow \mathbb{R}$ is continuous and $f(x) = f(x^p)$. Then

- (A.) f is constant if $p > 1$.
- (B.) f may not be constant for some $p > 1$
- (C.) f is constant if $p < 1$.
- (D.) None of the above is true.

TRUE:	A				FALSE:	B	C	D	
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12. Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$. Then

- (A.) If $A = (a, b)$ and f is continuous and bounded above, then there is $c \in (a, b)$ such that

$$f(c) = \sup_{x \in (a, b)} f(x).$$

- (B.) If $A = \cup_{n=1}^{\infty} [n - 1/2, n + 1/2]$ and f is continuous on A then f is bounded on A .
- (C.) If $A = [a, b]$ then f is continuous implies the intermediate value theorem hold.
- (D.) If $A = [1, 2] \cup [3, 4]$ and range of f is A then there is a c in A such that $f(c) = c$.

TRUE:	C				FALSE:	A	B	D	
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13. The following functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are differentiable at 0

(A.) $f(x) = |x|$.

(B.) $f(x) = |x|^3$.

(C.) $f(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$

(D.) $f = g^{-1}$ where $g(x) = x^3 + 2x + 3$.

TRUE:	B	C	D		FALSE:	A			
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14. If we expand the polynomial $p(x) = x^4 + x^2 + x + 1$ in terms of the power of $(x - 1)$, then

(A.) The constant term in this expansion is 4.

(B.) The constant term in this expansion is 5.

(C.) The coefficient of $(x - 1)^2$ in this expansion is 7.

(D.) The coefficient of $(x - 1)^3$ in this expansion is 5.

TRUE:	A	C			FALSE:	B	D		
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